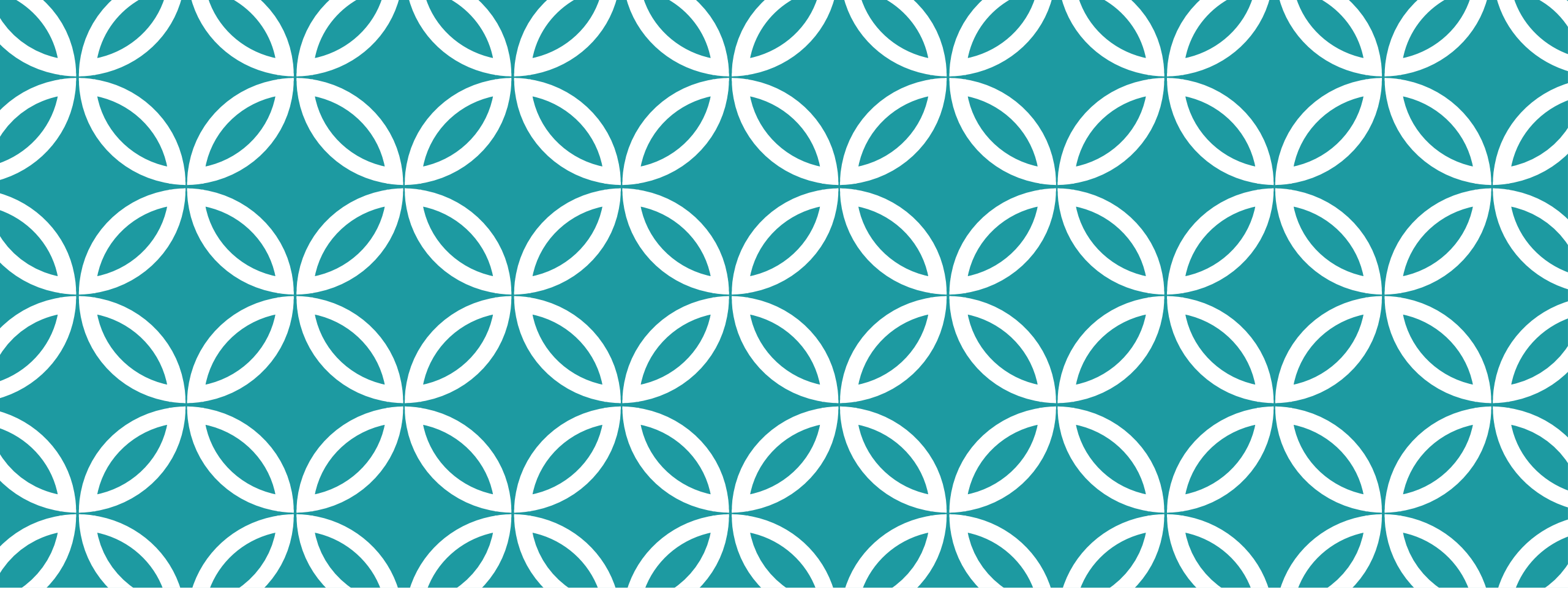


PUBP 233
INTRODUCTION TO DEMOGRAPHY
(POPULATION AND SOCIETY)

**Week 4: Introduction to Formal
Demography**

Shivakumar Jolad



FORMAL DEMOGRAPHY |

JOHN GRAUNT



CAPTAIN JOHN GRAUNT

Natural and Political OBSERVATIONS

Mentioned in a following INDEX,
and made upon the
Bills of Mortality.

By JOHN GRAUNT,
Citizen of
LONDON.

With reference to the Government, Religion, Trade,
Growth, Age, Disease, and the several Changes of the
said CITY.

— *Nas, ut arripitur Turba, labora.*
Contentus pavori LeBarbar —

LONDON,

Printed by Tho: Streater, for John Martin, James Allcock,
and Tho: Dine, at the Sign of the Bell in St. Paul's
Church-yard, MDCLXII.

1606. A TABLE of the CHRISTENINGS and MORTALITY

For the Year 1605 and 1606.*

Weeks.	Days of the Month.	Christ.	Bur.	Pla.	Par. infec.	Weeks.	Days of the Month.	Christ.	Bur.	Pla.	Par. infec.
1	Dec. 26.	100	116	5	5	28	July 3.	109	110	25	12
2	January 2.	117	151	6	5	29	10.	111	134	33	18
3	9.	130	138	4	4	30	17.	115	146	50	22
4	16.	124	138	3	2	31	24.	96	140	46	26
5	23.	143	121	6	4	32	31.	132	178	66	29
6	30.	124	101	3	2	33	August 7.	131	181	67	29
7	Febr. 6.	122	105	5	5	34	14.	141	197	75	33
8	13.	131	118	7	6	35	21.	133	189	85	28
9	20.	126	109	12	6	36	28.	125	207	85	29
10	27.	102	117	9	8	37	Septem. 4.	123	241	116	32
11	March 6.	110	98	7	4	38	11.	134	216	105	28
12	13.	126	137	9	7	39	18.	121	214	92	36
13	20.	123	133	14	11	40	25.	132	204	87	35
14	27.	134	123	17	8	41	October 2.	121	256	141	40
15	April 3.	123	114	13	9	42	9.	134	218	106	38
16	10.	132	145	27	11	43	16.	142	227	117	37
17	17.	139	129	12	8	44	23.	131	224	109	38
18	24.	118	110	11	7	45	30.	124	226	101	34
19	May 1.	92	136	17	10	46	Novem. 6.	136	183	68	27
20	8.	116	103	13	11	47	13.	125	162	41	20
21	15.	128	94	13	8	48	20.	121	145	28	11
22	22.	113	132	14	9	49	27.	143	123	22	13
23	29.	94	98	9	7	50	Decem. 4.	155	160	45	17
24	June 5.	129	112	16	8	51	11.	135	137	38	20
25	12.	127	112	19	14	52	18.	136	132	28	15
26	19.	121	119	15	10	53	25.	134	135	38	19
27	26.	132	126	24	16						

The Totals { Christened — — 6614
Buried — — 7920
Whereof of the Plague 2124

* BELL's London's Remembrancer.

166
2

Source: wiki-
http://it.wikipedia.org/wiki/John_Graunt

- Cause of Death
- Proportion of Surviving
- Family size
- Age structure

Epidemiology

Public Health

History of Statistics 1

The Bills of Mortality, and the Beginning of Statistics. John Graunt (1620 –1674)

From ancient times people have counted stuff: money, cattle, taxpayers and soldiers, to name a few. It is impossible to pick a date to mark the beginning of “statistics,” as used in this general sense. For our purposes it is reasonable to choose a time when the use of numerical information became noticeably similar to current approaches.

Beginning in the middle 1500s, during the reign of Henry VIII, parish registrars in England began to keep records of all weddings, christenings and burials for all parishioners in the Church of England. The records were used especially to keep track of deaths caused by epidemics of bubonic plague. Beginning in 1603 such lists – called *bills* - were regularly printed for the public. By 1625 the enterprise was well established with a dedicated printing press under the offices of the Archbishop of Canterbury. These *bills of mortality* were published weekly. As you can see in Figure 1, they include some strange causes of death.*

In about 1660 **John Graunt**, a respected businessman in London, began to analyze the whole collection of bills. In 1662 he published a small book, *Natural and Political Observations on the Bills of Mortality* based on about 70 years of data. As you can see from the first words of the preface to the book, he thought that the information in the bills was good for more than just gossip or quick business opportunities. As a result, he is considered a founder of two fields that make heavy use of statistics, *demography* (descriptions of populations) and *epidemiology* (study of diseases in populations).

Fig. 1 A Bill of Mortality from 1665

The Diseases and Casualties this Week.			
		Imposthume	11
		Infants	16
		Killed by a fall from the Bell-frey at Alhallows the Great	1
		Kingsevil	2
		Lethargy	1
		Pallie	1
		Plague	7165
		Rickets	17
		Rising of the Lights	11
		Scowring	5
		Scurvy	2
		Spleen	1
		Spotted Feaver	101
		Stillborn	17
		Stone	2
		Stopping of the Stomach	2
		Strangury	1
		Suddenly	1
		Surfeit	49
		Teeth	121
		Thrush	5
		Timpany	1
		Tiffick	11
		Vomiting	3
		Winde	3
		Wormes	15
A Borrive		5	
Aged		43	
Ague		2	
Apoplexie		1	
Bleeding		2	
Burnt in his Bed by a Candle at St. Giles Cripplegate		1	
Canker		1	
Childbed		42	
Chrisomes		18	
Consumption		134	
Convulsion		64	
Cough		2	
Dropfie		33	
Feaver		309	
Flox and Small-pox		5	
Frighted		3	
Gowr		1	
Grief		3	
Gripping in the Guts		51	
Jaundies		5	
Christened		Males — 95 Females — 81 In all — 176	Buried { Males — 4093 Females — 4202 In all — 8297 } Plague — 7165
Increased in the Burials this Week		607	
Parishes clear of the Plague		4	Parishes Infected — 126
The Assize of Bread set forth by Order of the Lord Mayor and Councill of Aldermen, A penny Wheaten Loaf to contain Nine Ounces and a half, and three half-penny White Loaves the like weight.			

Graunt's Life table

9. Whereas we have found, that of 100 quick Conceptions about 36 of them die before they be six years old, and that perhaps but one surviveth 76, we, having seven Decads between six and 76, we sought six mean proportional numbers between 64, the remainder, living at six years, and the one, which survives 76, and finde, that the numbers following are practically near enough to the truth; for men do not die in exact Proportions, nor in Fractions: from whence arises this Table following.

Viz. of 100 there dies within the first six years	36
The next ten years, or Decad	24
The second Decad	15
The third Decad	09
The fourth	6
The next	4
The next	3
The next	2
The next	1

10. From whence it follows, that of the said 100 conceived there remains alive at six years end 64.

At Sixteen years end	40
At Twenty six	25
At Tirty six	16
At Fourty six	10
At Fifty six	6
At Sixty six	3
At Seventy six	1
At Eighty	0

Graunt's life table in modern format

Age interval	Number of deaths in interval	Number alive the start of the interval	Proportion of deaths in interval	Proportion alive the start of the interval
0-6	36	100	.36	1.00
7-16	24	64	.24	.64
17-26	15	40	.15	.40
27-36	9	25	.09	.25
37-46	6	16	.06	.16
47-56	4	10	.04	.10
57-66	3	6	.03	.06
67-76	2	3	.02	.03
77-86	1	1	.01	.01

ALFRED LOTKA, FISHER, KEYFITZ



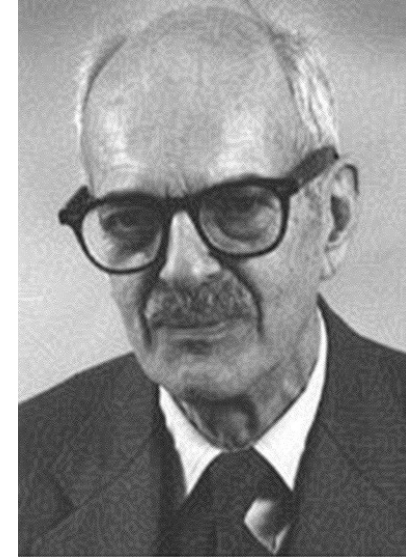
Alfred Lotka (1880–1949)

- Lotka's equation
- Stable population
- Predator Prey model etc.



Ronald Fisher (1890–1962)

- Evolutionary biologist
- Fisher equation, Fisher reproductive value etc



Nathan Keyfitz (1913–2010)

- Population Momentum
- Formal Demography
- Population Biology

POPULATION CHANGE

Births, Deaths and
Migration

$$P(t + 1) = P(t) + \text{Births} - \text{Deaths} + \text{In migration} - \text{Out migration}$$

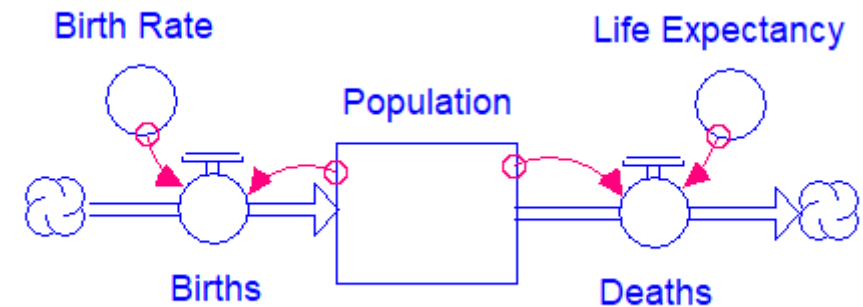
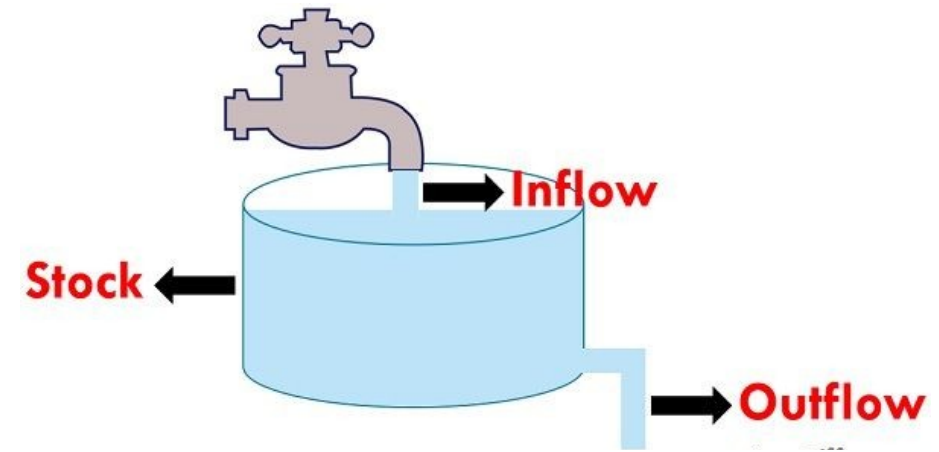
Every year people age by one
year!

Dynamics aspects : **Fertility, Mortality, Migration, Nuptiality, Age structure ...**

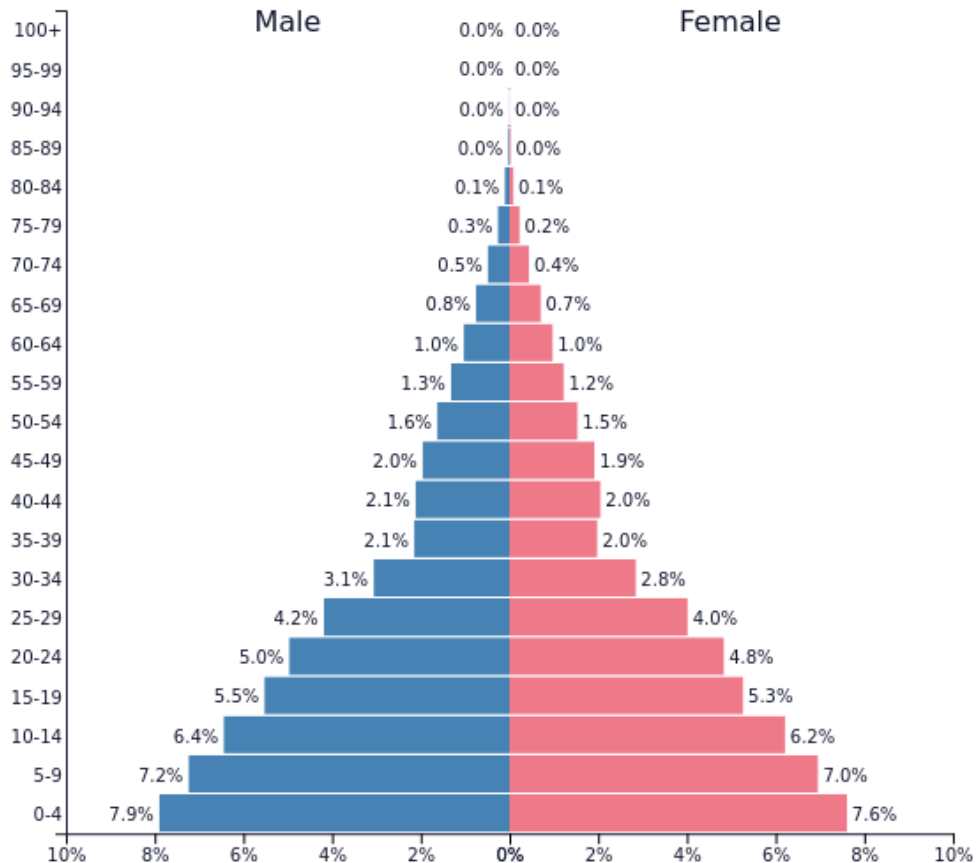
POPULATION STOCK AND FLOWS

Population stock is the size and composition of a population at a given moment in time. The population stock can be changed by population flows, and current population stock reflects past population flows.

Population flows are the demographic events of mortality, fertility and migration which bring about changes in the size and structure of a population.



AGE-SEX STRUCTURES AND POPULATION PYRAMIDS



The age-sex structure of a population is simply the distribution of the population by age and sex. It may be described in terms of absolute numbers or percentages.

Proportion of the population under a certain age (often 15) or over a certain age (often 65)

Median age of a population i.e. the age which half the population are younger and half the population is older

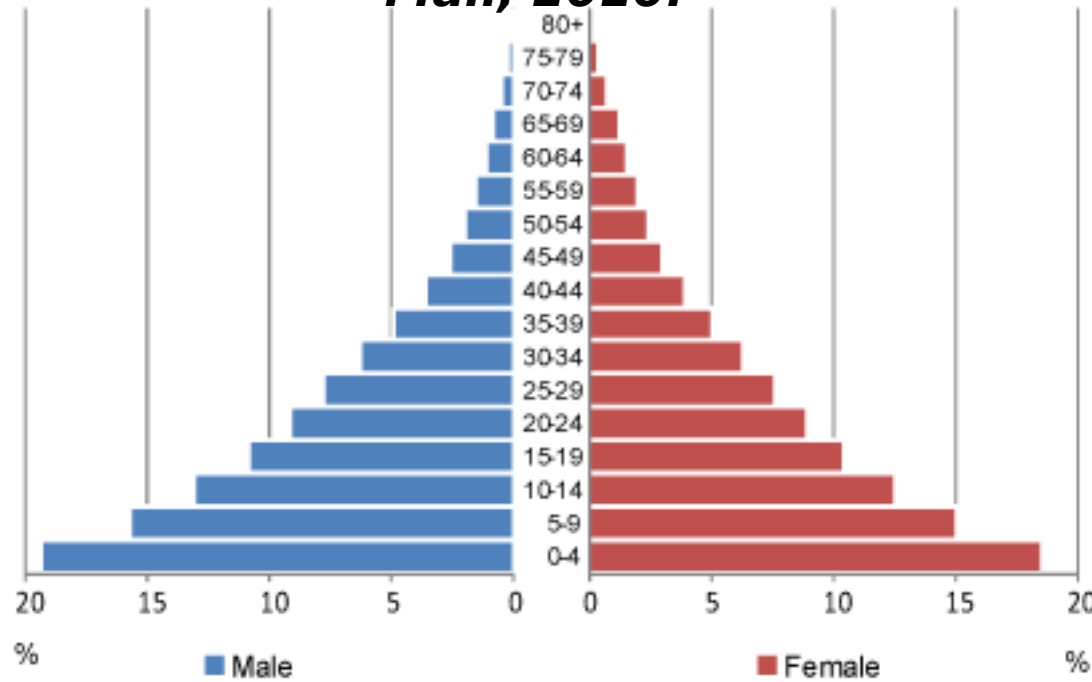
Dependency ratios

Child dependency ratio

Old age dependency ratio

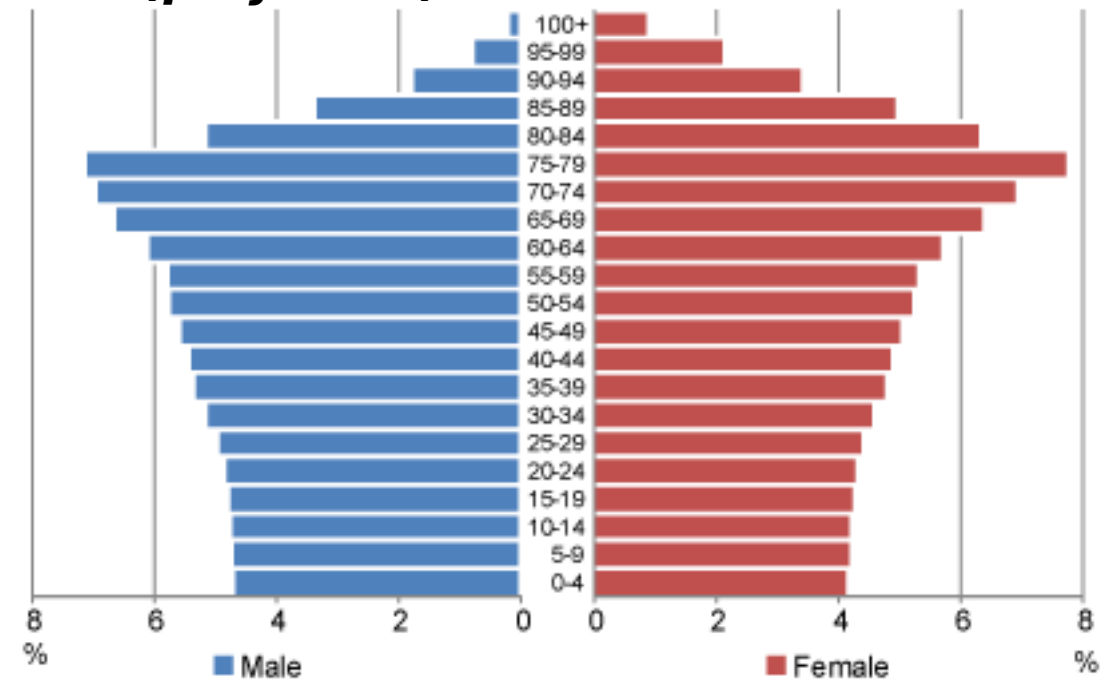
DESCRIBING POPULATION PYRAMIDS

Population pyramid for Mali, 2010.



Young and growing

Population pyramid for Japan, 2050 (projected).



Old and declining

http://papp.iussp.org/sessions/papp101_s02/PAPP101_s02_040_030.html

MEASURES TO DESCRIBE THE AGE STRUCTURE OF A POPULATION

Proportion of the population under a certain age (often 15) or over a certain age (often 65)

Median age of a population i.e. the age which half the population are younger and half the population is older

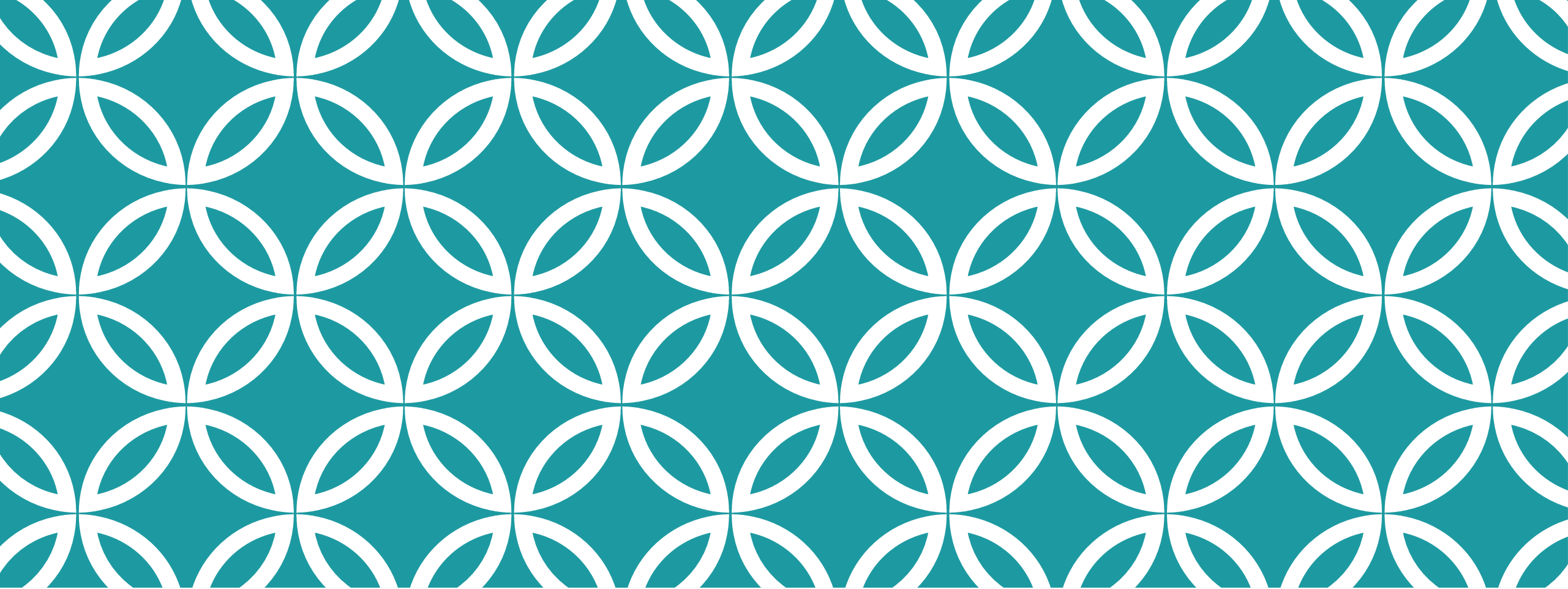
Dependency ratios

Child dependency ratio

Old age dependency ratio

Census 2011					
Age group	Persons	Males	Females	Sex ratio (M/F*100)	
0-4	112806778	58632074	54174704		
5-9	126928126	66300466	60627660		
10-14	132709212	69418835	63290377		
15-19	120526449	63982396	56544053		
20-24	111424222	57584693	53839529		
25-29	101413965	51344208	50069757		
30-34	88594951	44660674	43934277		
35-39	85140684	42919381	42221303		
40-44	72438112	37545386	34892726		
45-49	62318327	32138114	30180213		
50-54	49069254	25843266	23225988		
55-59	39146055	19456012	19690043		
60-64	37663707	18701749	18961958		
65-69	26454983	12944326	13510657		
70-74	19208842	9651499	9557343		
75-79	9232503	4490603	4741900		
80-84	6220229	2927040	3293189		
85+	5068776	2356655	2712121		
	1,20,63,65,175	62,08,97,377	58,54,67,798		

Questions		
	Calculate the sex ratio of Population 1 (Female/Males (per thousand)	
	Calculate Male to Female sex ratio in 2 each age group - per 100 ($M/F \times 100$)	
	Create a population Pyramid of India - describe the characteristics ; Highlight 3 peak	
	Mean age of Population, Median age of population (do by sex- is there a 4 gender difference?)	
	5 Child dependency ratio : 0-14 / 15-65	
	Old age dependency ratio : 65+ yrs / 615-64 years	



RATES AND PROBABILITIES |

RATES

$$\text{Rates } [0, T] = \frac{\text{No of occurrences between 0 to T}}{\text{Person years lived between 0 to T}}$$

**Crude Birth
rate**

**Crude Death
Rate**

**Infant
Mortality Rate**

PERIOD RATES

$$CBR[0, T] = \frac{\text{Number of Births } [0, T]}{PY[0, T]}$$

$$CDR[0, T] = \frac{\text{Number of Deaths } [0, T]}{PY[0, T]}$$

$$CIM[0, T] = \frac{\text{Number of In migrations } [0, T]}{PY[0, T]}$$

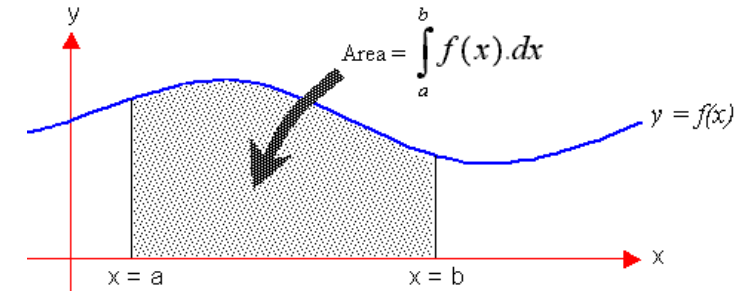
$$COM[0, T] = \frac{\text{Number of Out migrations } [0, T]}{PY[0, T]}$$

CALCULATION OF PERSON YEARS

Year	2000	2001	2002	2003	2004	Years at risk
Persons						
1	-----	-----	-----	-----	-----	5
2	-----	-----	-----	-----	-----X	4.5
3	-----	-----	-----	-----X		3.5
4		-----	-----X			1.5
5	-----	-----	-----	-----X		3.5
Total	4	5	4.5	3	1.5	18

$$PY[0, T] = \sum_{t=0}^T N(t) \times \Delta t$$

$$PY[0, T] = \int_0^T N(t) dt \approx \sum_{i=0}^T N_i \Delta_i \approx \frac{(N(0) + N(T))}{2} T$$



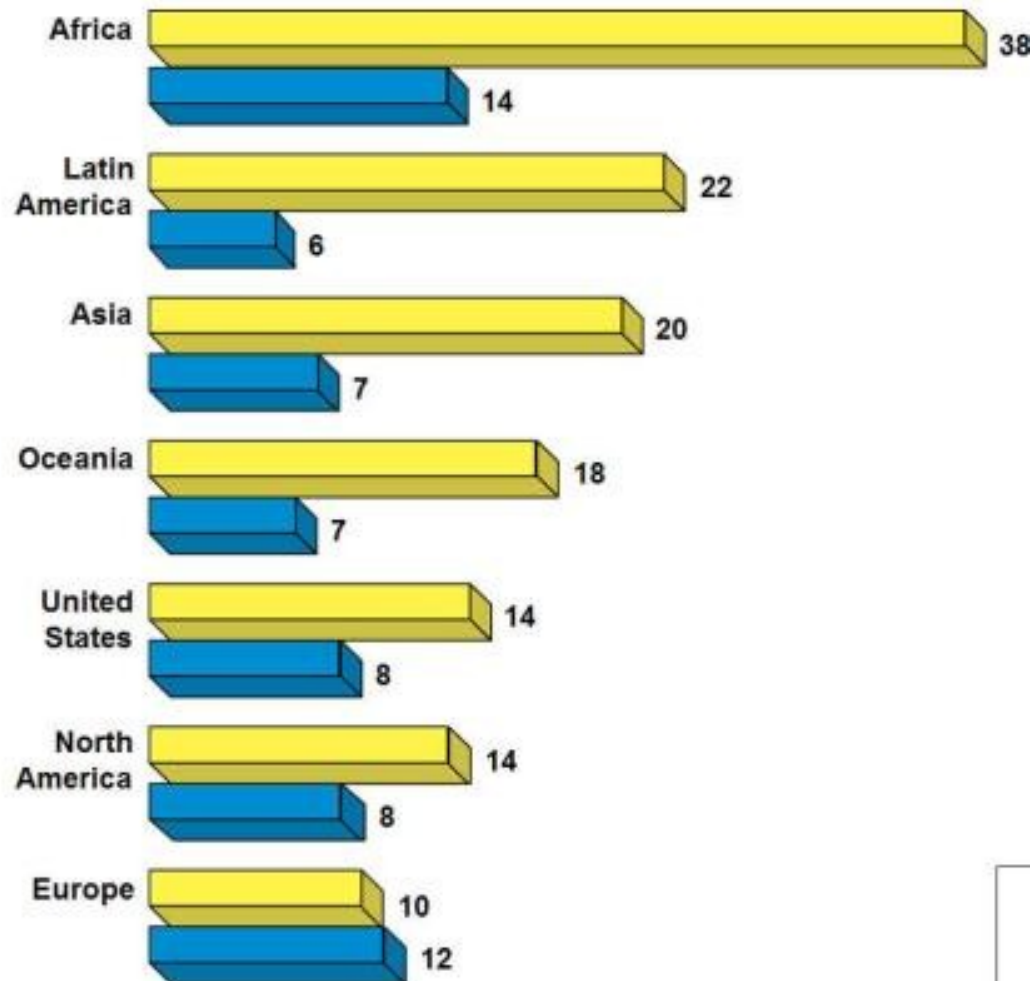
CDR

The crude death rate (CDR) is the simplest and one of the more common indicators of mortality in a population. It is the ratio of the numbers of deaths (D) observed in a population in a year to the population at risk of dying in that year (N), usually multiplied by 1000:

$$CDR = \frac{D}{N} \cdot 1000$$

Demographers tend to use a period of a year – typically a calendar year running from 1 January to 31 December – to derive a CDR so as to eliminate possible seasonal fluctuations in mortality (deaths of the elderly, for example, tend to occur disproportionately in winter) while still retaining a degree of precision in the estimate by not smoothing out trends in mortality over a number of years.

CRUDE BIRTH AND DEATH RATES - WORLD

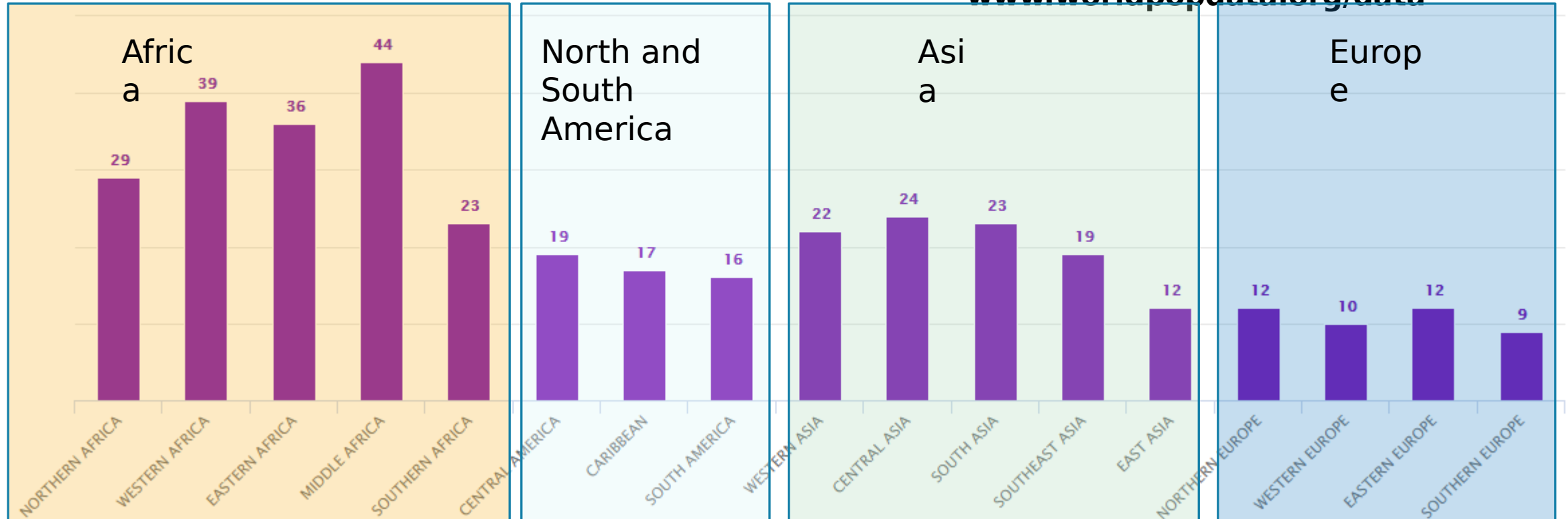


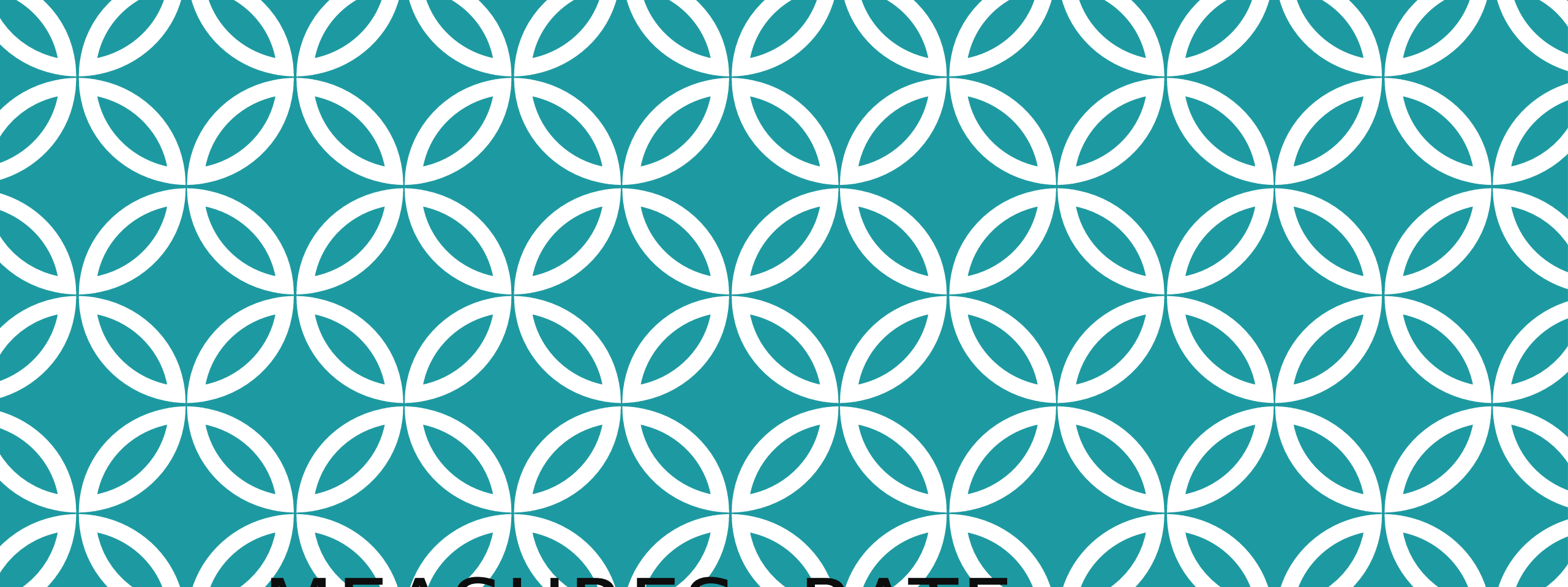
2014

CRUDE BIRTH RATES - WORLD

Births per 1,000 Population

<http://www.worldpopdata.org/data>





**MEASURES: RATE,
PREVALENCE,
INCIDENCE,
EXPOSURE**

RATES AND PREVALENCE

$$\text{Rates } [0,T] = \frac{\text{No of occurrences between 0 to T}}{\text{Person years lived between 0 to T}}$$

Prevalence is the proportion of the population with the outcome of interest at a given point in time. It is often used to assess the burden of disease within a population.

$$\text{prevalence} = \frac{\text{number of existing cases}}{\text{total population}} \text{ at a given point in time}$$

Example: (Point
Prevalence)

In 2002, HIV tests were given to consenting women aged 25 or less, attending antenatal clinics in the Mwanza region of Tanzania. Results were divided into four categories based on the location of the clinics: Rural – remote; Rural – roadside; Urban – town; and Urban – city.

In the Rural – roadside setting, 1042 women were tested. Of these, 107 were found to be HIV+.

From this we can calculate the prevalence:

$$\text{HIV prevalence} = \frac{107}{1042} \times 100 = 10.3\%$$

Period Prevalence - Measured over a

SEROPREVALENCE

Pointers from serosurveys conducted in various cities

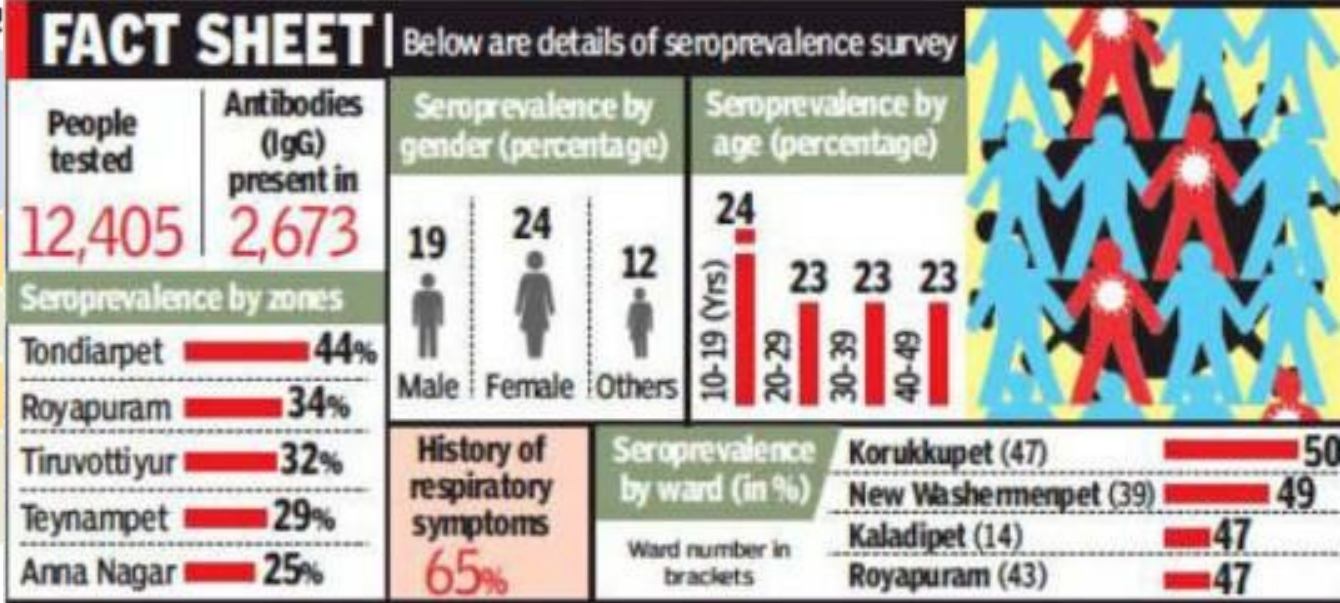
One-fifth of Chennai infected with Covid-19 till mid-July: Survey

Estimate of seropositivity across age groups

TABLE 2

	Number	Prevalence of seropositivity (in per cent)	95% CI
18-30 years	395	52.5	47.6-57.4
31-50 years	680	52.1	48.4-55.9
51-65 years	418	54.8	50.0-59.6
66 years and above	171	39.8	32.6-47.1

Source: IISER, Pune

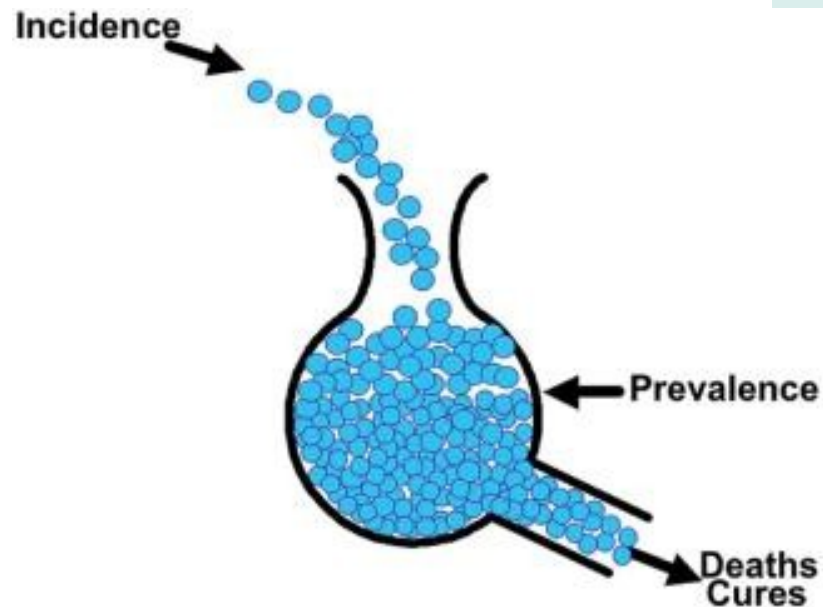


<https://frontline.thehindu.com/the-nation/sero-survey-pointers/article32419997.ece>
<https://timesofindia.indiatimes.com/city/chennai/one-fifth-of-chennai-infected-with-covid-19-till-mid-july-survey/articleshow/77881551.cms>

INCIDENCE

Incidence quantifies the **number of new events** which occur in a population during a specific time period.

$$\text{incidence} = \frac{\text{number of new cases}}{\text{total population}} \text{ during the specified time period}$$



Eg:
COVID Incidence Rate=
$$\frac{\text{New incidences of COVID in Pune}}{\text{Total Population of Pune}}$$

http://papp.iussp.org/sessions/papp101_s02/PAPP101_s02_080_010.html

WORLD POPULATION PROSPECTS -2024- DATA PORTAL



United
Nations

Data Portal
Population Division

Home

Data

Country Profiles

Tools ▼

About ▼

UN Population Division Data Portal

Interactive access to global demographic indicators

Search data by indicator and location

6 Indicators selected



265 Locations selected



DATE RANGE

2025



–

2025



Get Started and Search

OR USE

▶ Selection Wizard

Indicator Options

Indicator

Classes

Settings

Palettes

INDICATORS

Child dependency ratio

Crude birth rate

Crude death rate

Median age of population

Old-age dependency ratio

Total dependency ratio

DIMENSIONS/CHARACTERISTICS

P.I.

0 / 3

Var

1 / 11

Median

Constant-fert.

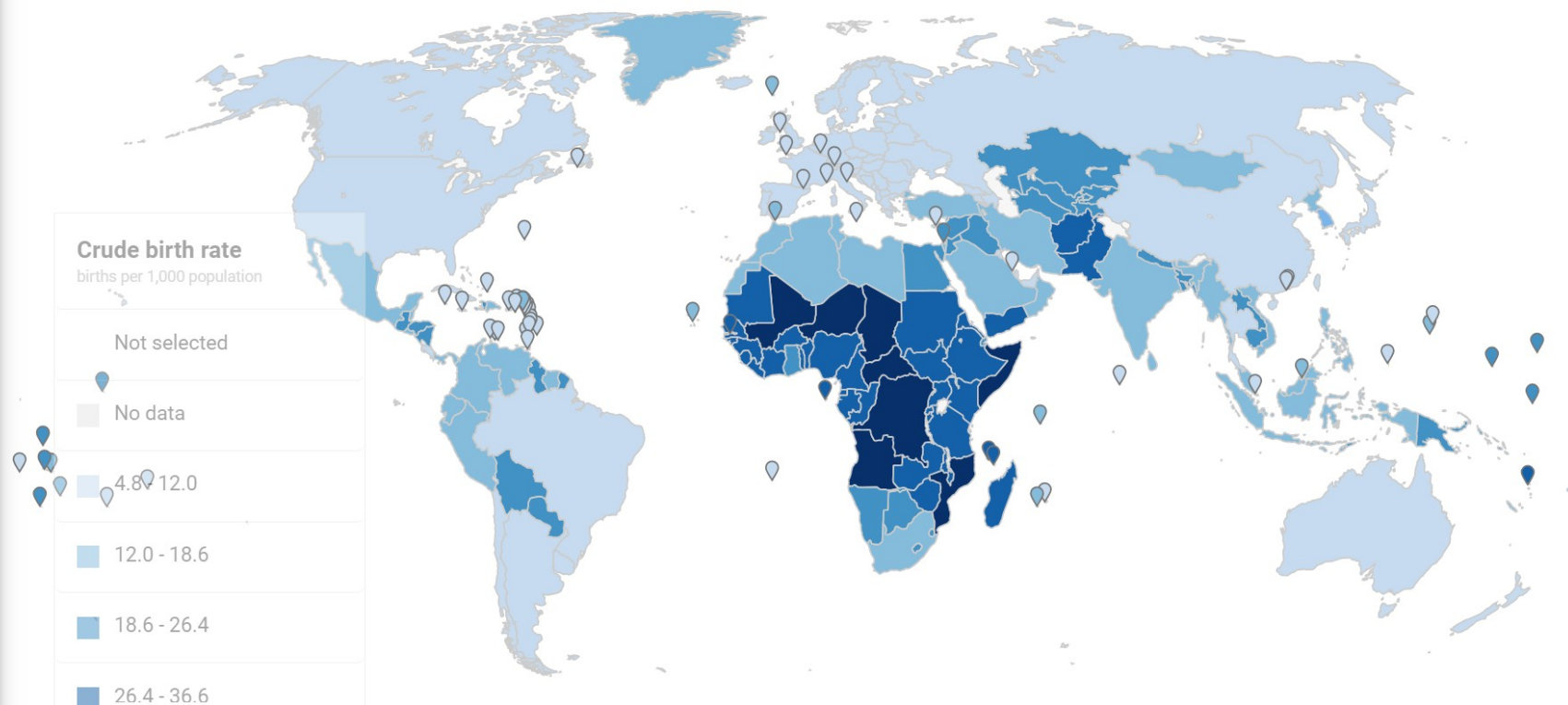
Inst.-replac.-fert.

Inst.-replac.-Zero-mig.

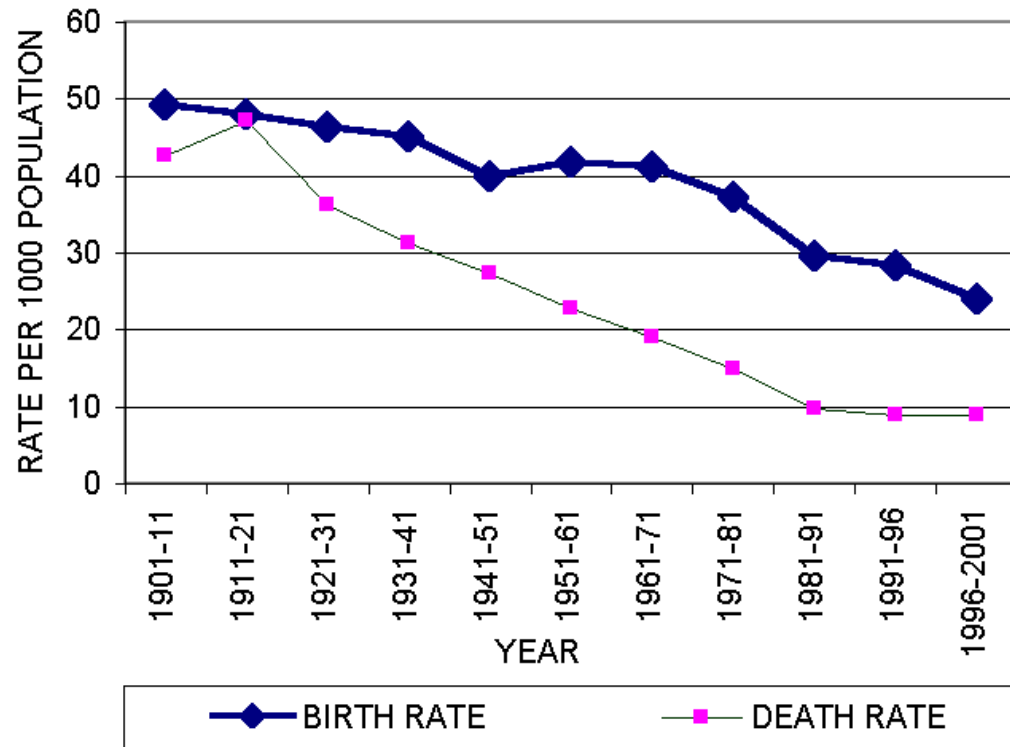
[Reset Default Dimensions](#)

2025

2025
Crude birth rate
Sex: Both sexes
births per 1,000 population

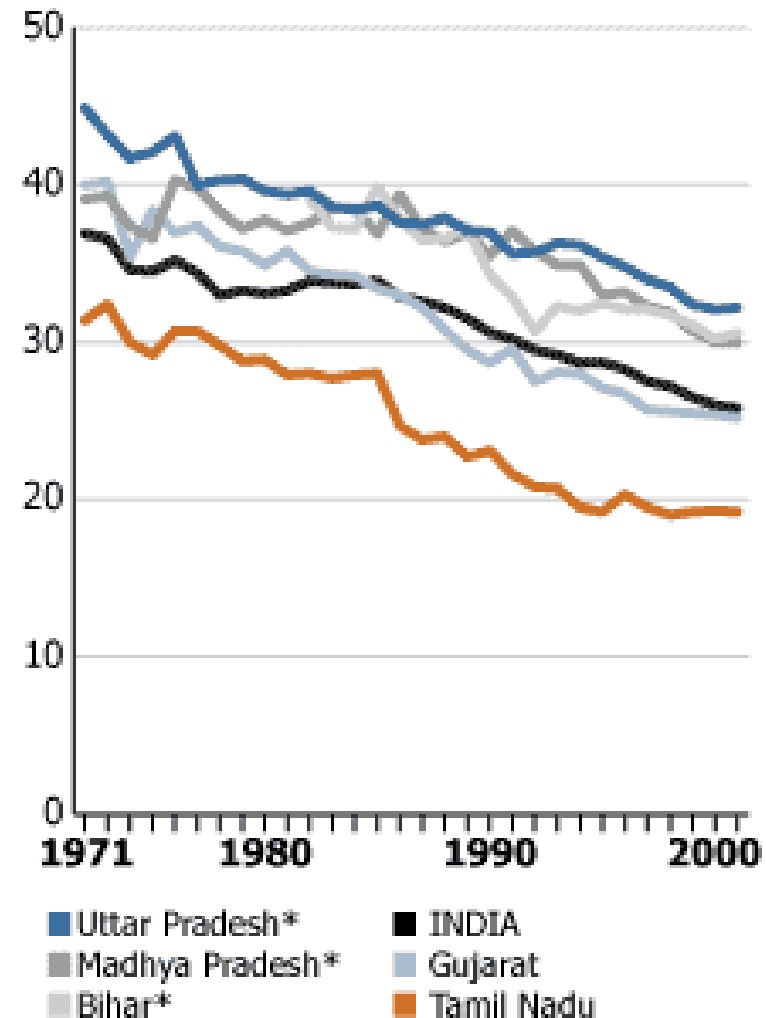


CBR AND CDR -



Source:- Registrar General India

INDIA Births per 1,000 population

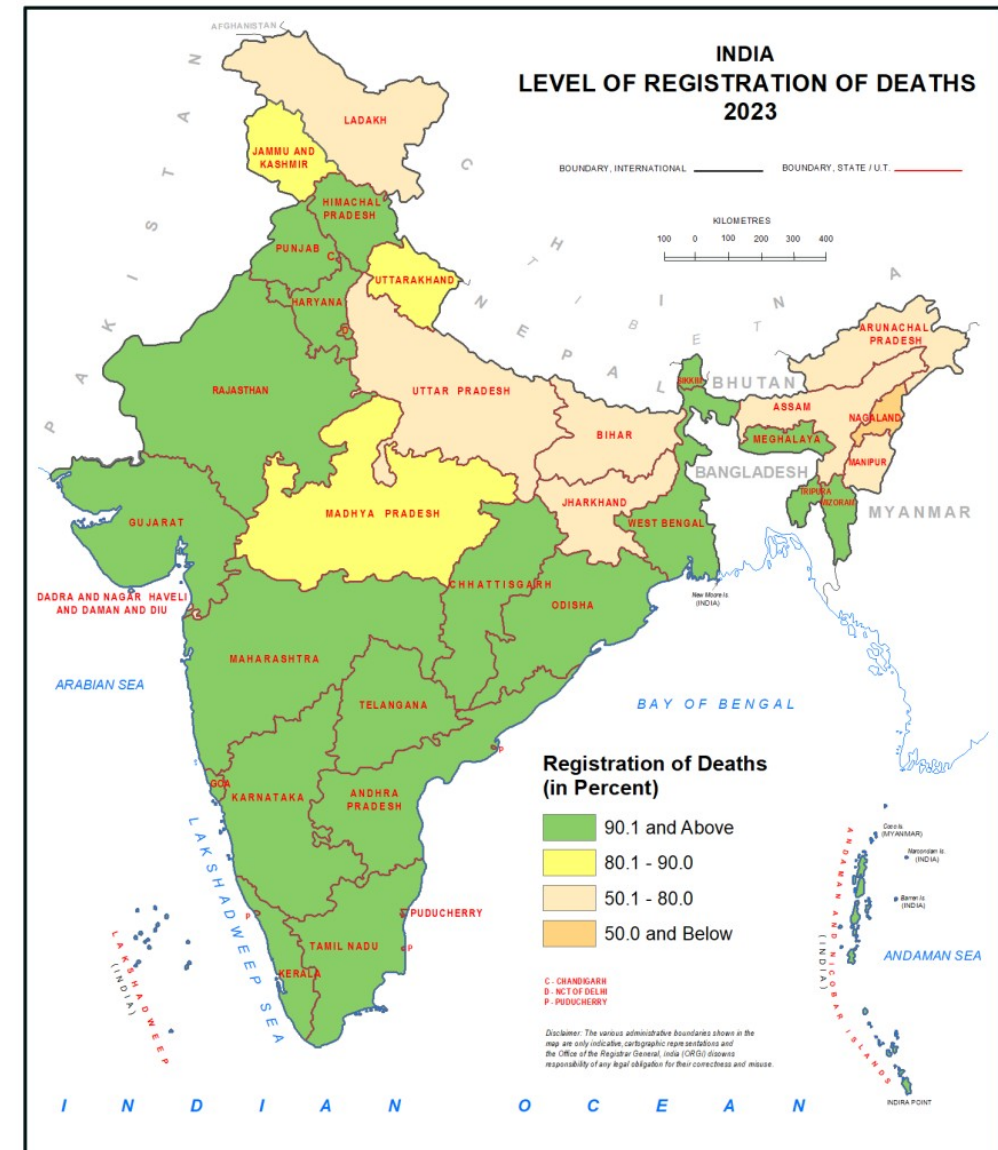
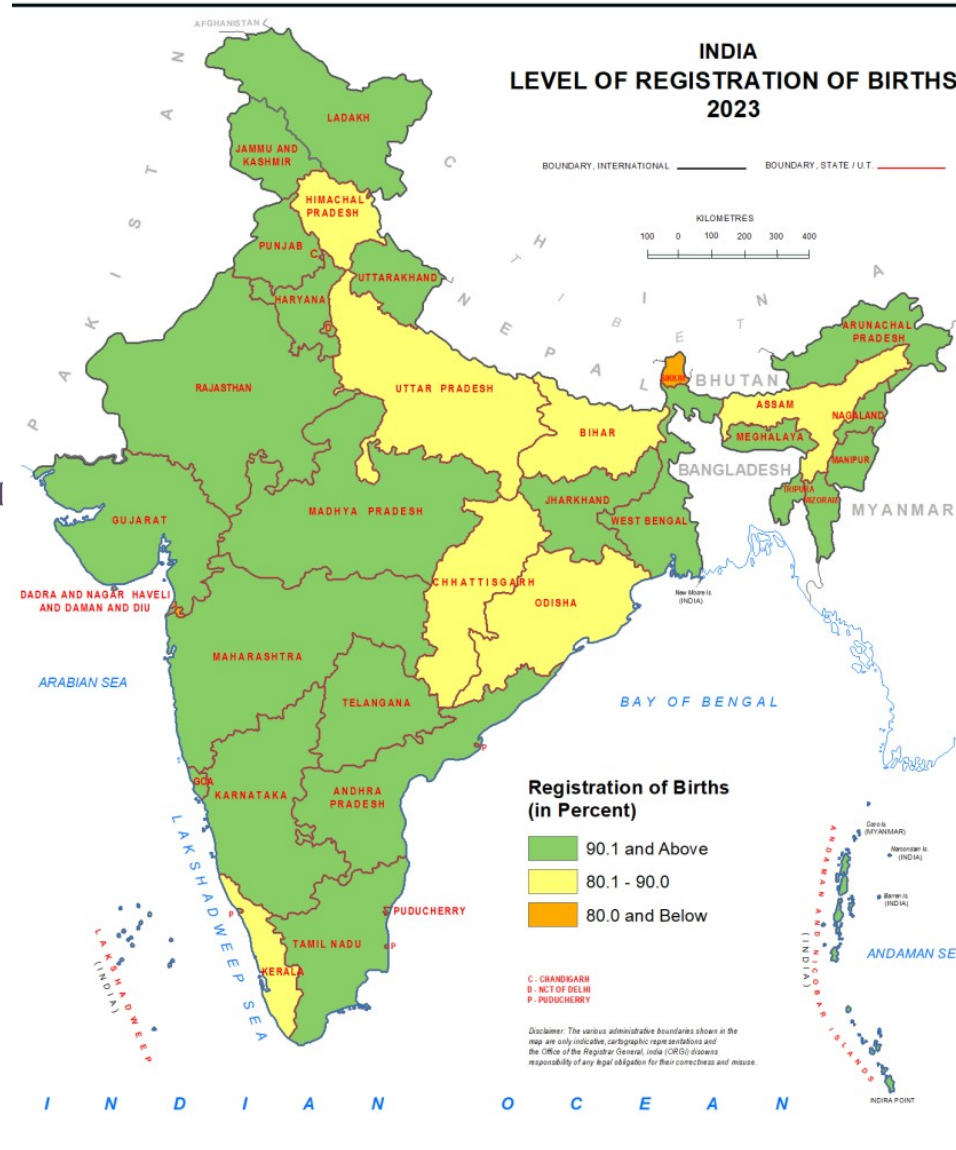


Carl Haub, PRB
publications, 2002

INDIA- CIVIL REGISTRATION SYSTEM



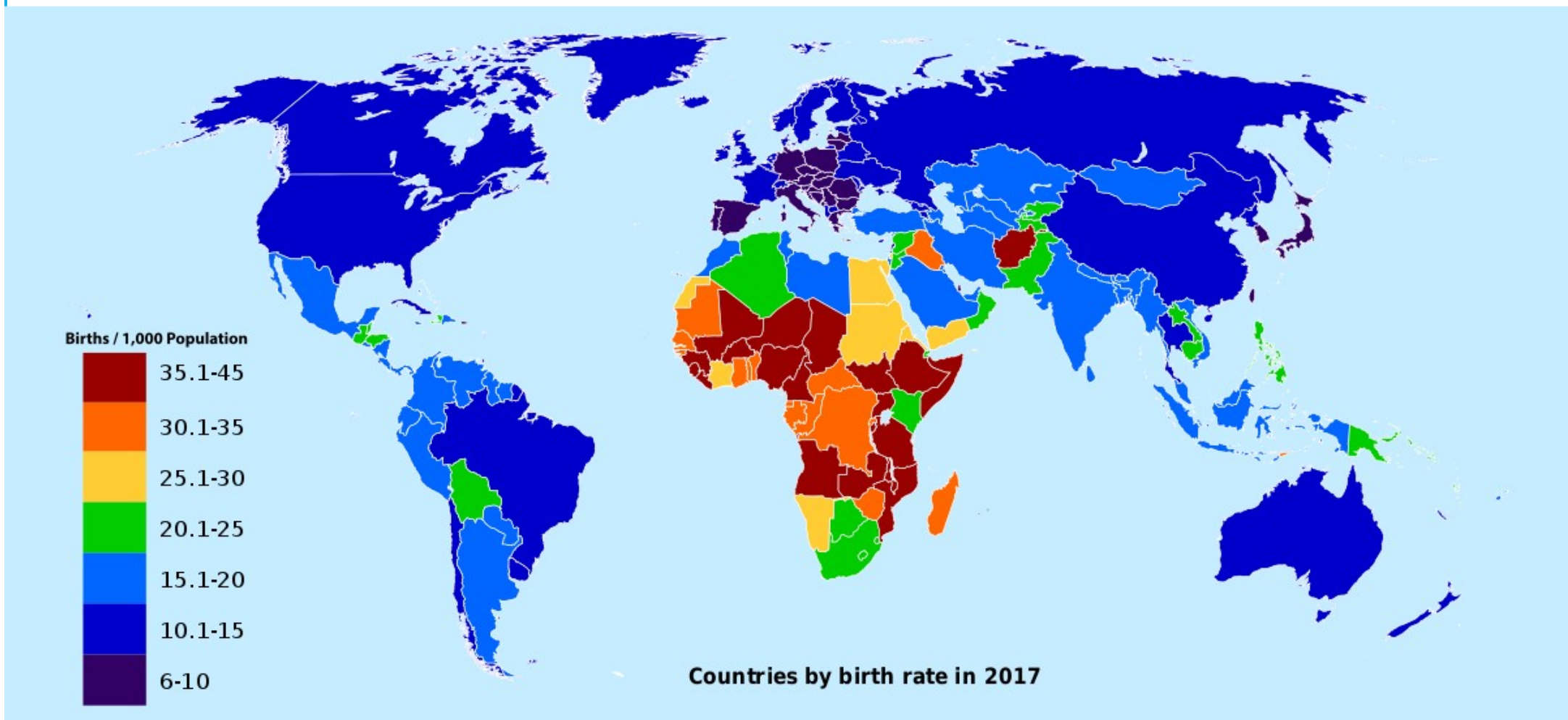
VITAL STATISTICS OF INDIA BASED ON THE CIVIL REGISTRATION SYSTEM 2023



CRS- STATS 2016

Statement 1: Civil Registration System: At A Glance

Demographic features of India as per the Civil Registration System (CRS)			
		2015	2016
Registration of Births			
i)	Estimated Mid- Year Population (in '000)	12,59,108	12,73,986
ii)	Estimated Live Births	2,61,89,446	2,59,89,314
iii)	Number of Registered Births#	2,31,36,145	2,22,00,991
iv)	Level of Registration	88.3	86.0
Registration of Deaths			
i)	Estimated Deaths	81,84,202	81,53,510
ii)	Number of Registered Deaths#	62,67,685	63,49,259
iii)	Level of Registration	76.6	78.1



What might be a good approximation to the population exposed to risk in the calculation of a crude death rate?

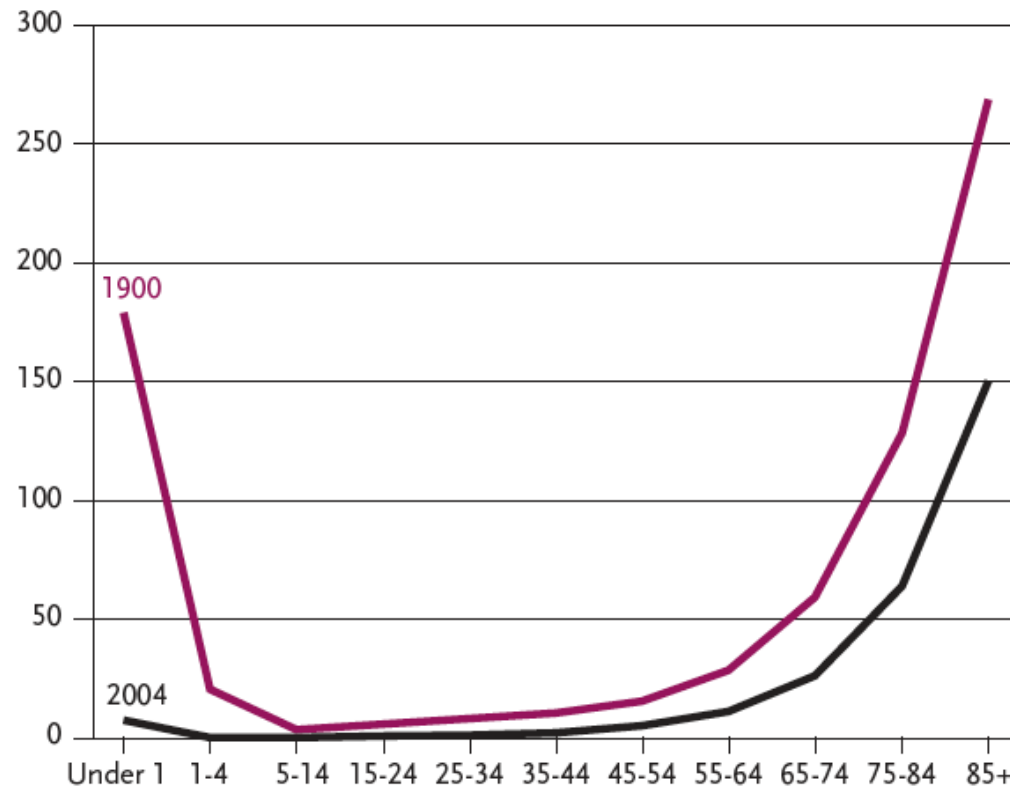
- a) Population at the start of the year
- b) Population at the middle of the year
- c) Population at the end of the year

Age Interval	Age (start)	Female	Male	Total	Deaths_Female	Deaths_Male	Deaths_Total
[0-1)	0	384383	402876	787259	1589	1915	3504
[1-4]	1	1486810	1557668	3044477	255	306	561
[5-9	5	1692231	1774079	3466310	145	181	326
[10-14]	10	1813124	1906552	3719676	169	198	367
15-19	15	1963965	2045365	4009330	412	741	1153
20-24	20	2086311	2088253	4174564	481	1330	1811
25-29	25	2141406	2120883	4262288	665	1456	2121
30-34	30	2001255	2001712	4002967	923	1888	2811
35-39	35	2189682	2152107	4341789	1578	2727	4305
40-44	40	2360900	2314331	4675231	2631	4270	6901
45-49	45	2296231	2247166	4543397	3911	5988	9899
50-54	50	1993476	1963789	3957265	5620	8132	13752
55-59	55	1824525	1779348	3603873	7815	11753	19568
60-64	60	1911204	1843078	3754282	12551	18834	31385
65-69	65	1494886	1400847	2895733	15631	23092	38723
70-74	70	1297668	1156762	2454430	22574	30960	53534
75-79	75	1103656	888420	1992075	33570	39861	73431
80-84	80	873266	595564	1468829	49189	46609	95798
85-89	85	614791	322235	937026	61538	42819	104357
90-94	90	229168	85675	314842	43783	20726	64509

DEATH RATES BY AGE

Death Rates for U.S. Males, by Age, 1900 and 2004

Deaths per 1,000 males

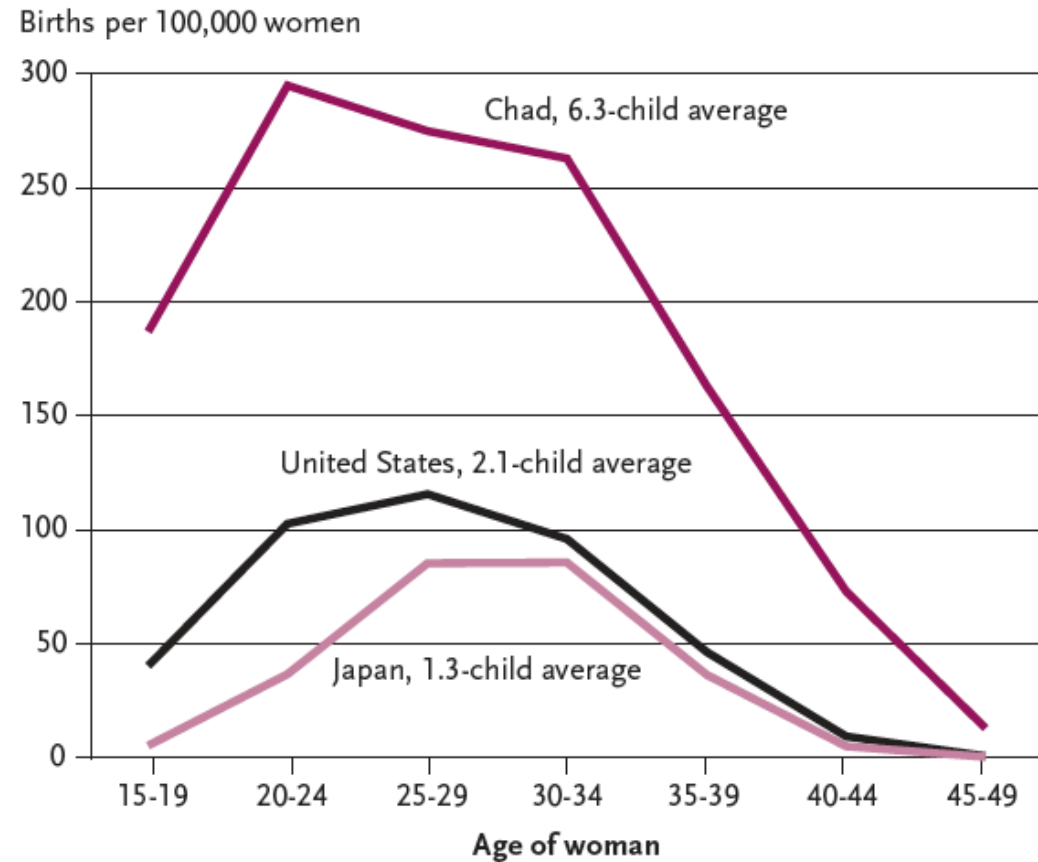


Source: U.S. Bureau of the Census, *Sixteenth Census of the United States: 1940, Vital Statistics Rates in the United States 1900-1940*, (1943): table 5; and National Center for Health Statistics, National Vital Statistics System, *Mortality, 2004* (forthcoming).

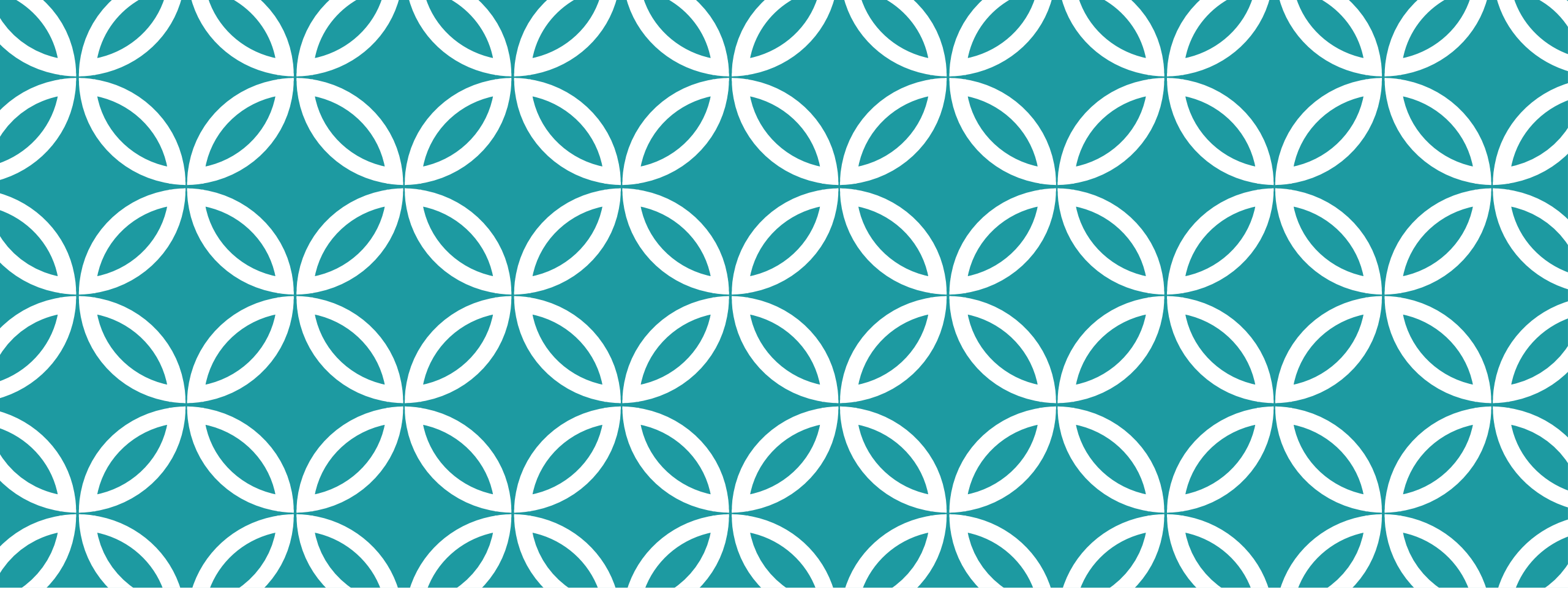
McFalls, Joseph A. *Population: a lively introduction*. (PRB, 2007)

AGE SPECIFIC FERTILITY RATE

Age Patterns of Fertility: Chad, Japan, and the United States, 2005



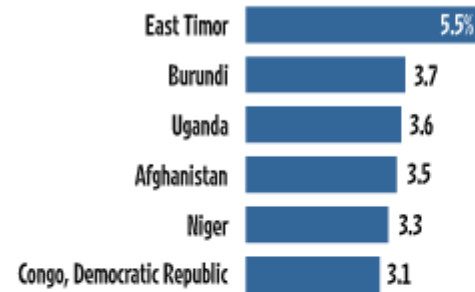
McFalls, Joseph A. *Population: a lively introduction*.



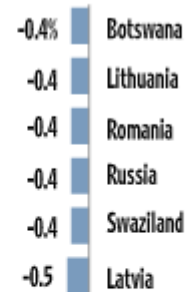
POPULATION GROWTH RATE |

POPULATION GROWTH RATE- WORLD

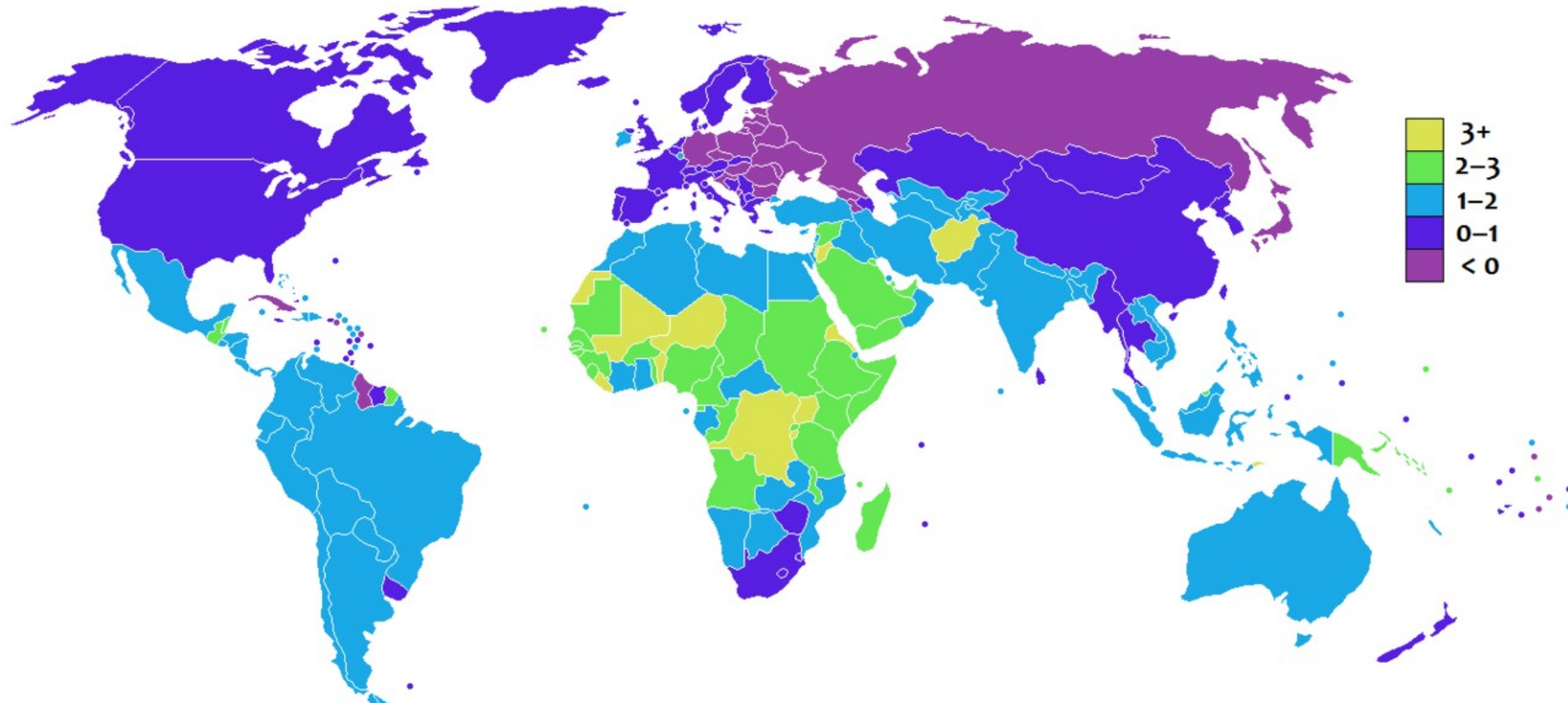
Fastest-growing populations



Slowest-growing populations



Source: Wall Street Journal
and wiki



POPULATION GROWTH RATES

$$N(t) = N_0 e^{rt}$$

A population with a growth rate of **1% per annum** will double in size **every 70 years**.

A growth rate of **2% per annum** will double in size every **35 years**.

$$T_d = \frac{\ln 2}{r} = \frac{0.693}{r}$$

A a growth rate of **3% per annum** will double in size every **23 years**.

India Annual growth rate (2021): 0.8%
(World Bank)

$$r(t) = \frac{N(t) - N(0)}{N(0) * t}$$

WHAT IF GROWTH REMAINED CONSTANT?

If the population growth remained at 2% annually :

- Less than 700 years, there would be one person for every square foot
- In less than 1200 years, human population would outweigh the earth !
- In less than 6000 years, the mass of Humanity would form a sphere expanding at the speed of light

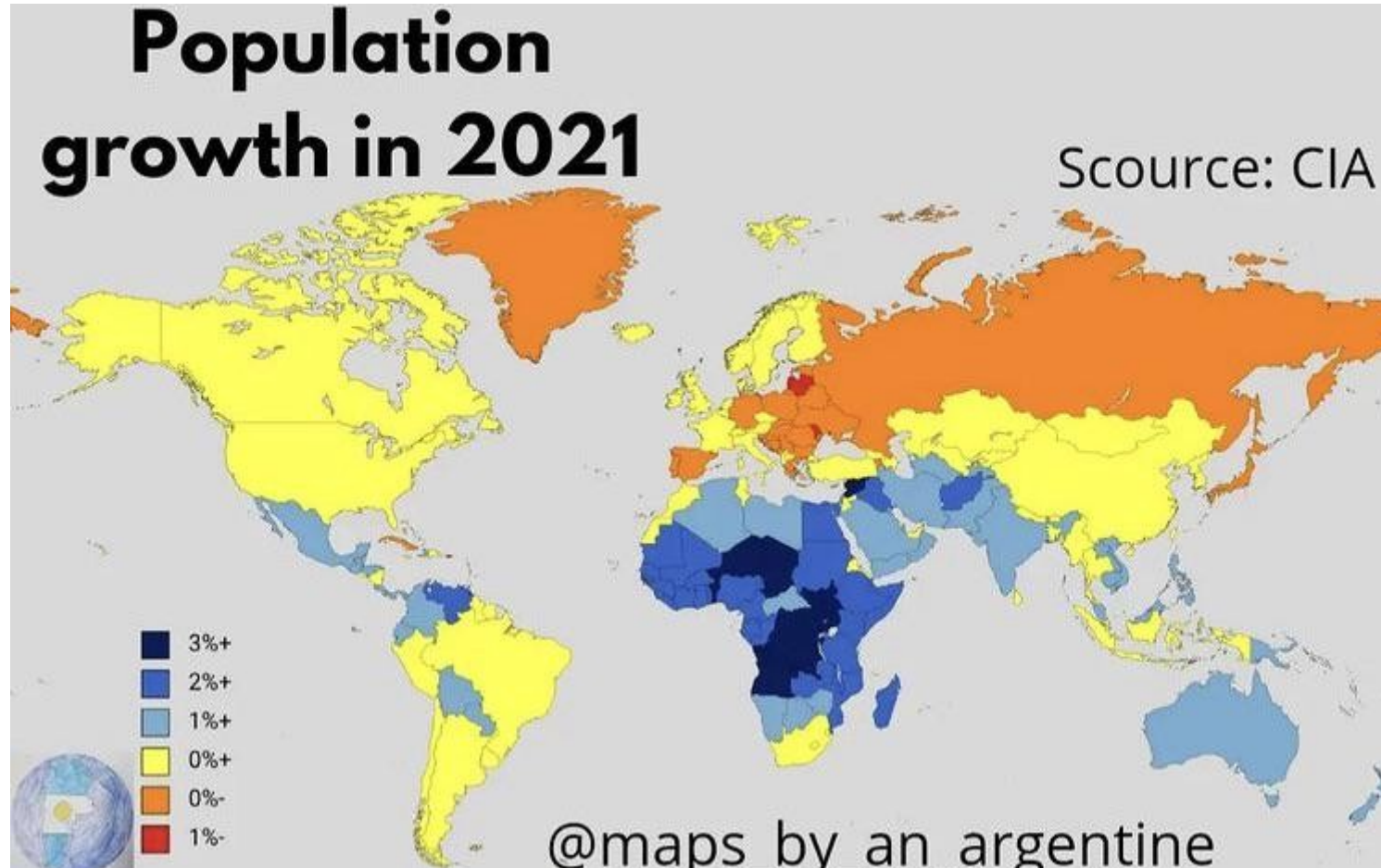


The Power of Geometric Growth

$$\frac{dN}{dt} = rN$$

$$N(t) = N_0 r^t$$

POPULATION GROWTH RATE- WORLD



CLASS QUIZ-

→ Select whether the given demographic process adds to a population, removes from a population or does neither. Click on the cells to do so.

	Adds	Removes	Neither
Adoption			
Emigration			
Fertility			
Immigration			
Marriage			
Mortality			

In 1990 a village had a population of 1000. Between 1990 and 2000 there were 160 births and 90 deaths in the village. 17 people moved into the village, and 23 people moved away.

https://papp.iussp.org/sessions/papp101_s01/

Based on these determinants of population ageing, and what you know about demographic trends in different areas of the world, **which region do you think has the highest proportions of the population in the older age groups?**

- a) Africa
- b) Asia
- c) Europe
- d) Latin America
- e) North America

2. Which of these regions do you think has the lowest proportion of the population in the older age groups?

- a) Africa
- b) Asia
- c) Europe
- d) Latin America
- e) North America

Do you think policies aimed at increasing fertility will lessen the consequences of population ageing in the short or long term?

- a) Short-term
- b) Long-term
- c) Both